

Inverse Reinforcement Learning: IRL aims to learn a reward function from the demonstrations and then learn a (near-)optimal policy from the learned reward.

* Consider a finite-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, H)$ and an expert (near-optimal) policy π^* .

A data set of demonstrations from π^* over $\mathcal{M}$ is available



$$\mathcal{D} = \{(s_i, a_i)\}_{i=1}^N \quad \text{where} \quad (s_i, a_i) \sim \bar{\gamma}_{\pi^*}^{s_0}$$

The objective is to learn a reward function such that π^* is an optimal policy with respect to that.



Assume that the transition function is known while the reward function is unknown but belongs to a parametrized class of

functions $R \in \{R_\theta : \theta \in \mathbb{R}^k\}$, for instance, a class of

linear functions $R \in \{R_\theta : R_\theta(s, a) = \theta^T \phi(s, a),$

$$\forall s \in \mathcal{S}, \forall a \in \mathcal{A}, \theta \in \mathbb{R}^k\}.$$

If π^* is an optimal policy with respect to the true reward

$$R = R_{\theta^*} \quad \text{where} \quad R_{\theta^*}(s, a) = \theta^{*T} \phi(s, a) \quad \forall s \in S, \forall a \in A,$$

then
$$v^{\pi^*}(\mu_0) \geq v^{\pi}(\mu_0) \quad \forall \pi \in \Pi$$

$$\rightarrow \mathbb{E}_{\mu_0, \pi^*, p} \left[\sum_{t=0}^T \gamma^t R_{\theta^*}(s_t, A_t) \right] \geq \mathbb{E}_{\mu_0, \pi, p} \left[\sum_{t=0}^T \gamma^t R_{\theta^*}(s_t, A_t) \right] \quad \forall \pi \in \Pi$$

$$\mathbb{E}_{\mu_0, \pi, p} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, A_t) \right] = \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi}} [f(s, a)]$$

$$\downarrow \rightarrow \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [R_{\theta^*}(s, a)] \geq \frac{1}{1-\gamma} \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi}} [R_{\theta^*}(s, a)] \quad \forall \pi \in \Pi$$

$$\rightarrow \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [\theta^{*T} \phi(s, a)] \geq \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi}} [\theta^{*T} \phi(s, a)] \quad \forall \pi \in \Pi$$

$$\rightarrow \theta^{*T} \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [\phi(s, a)] \geq \theta^{*T} \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi}} [\phi(s, a)] \quad \forall \pi \in \Pi$$

$\Rightarrow$ Therefore, if we find a policy π such that

$$\mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [\phi(s, a)] = \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi}} [\phi(s, a)],$$

policy π will attain the optimal value, i.e.,

$$v^{\pi^*} = v^{\pi}.$$

Idea: We can estimate the expected feature based on the data set.

$$\mathbb{E}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [\phi(s, a)] \approx \hat{\mathbb{E}}_{(s,a) \sim \bar{\gamma}_{\mu_0}^{\pi^*}} [\phi(s, a)] = \frac{1}{N} \sum_{i=1}^N \phi(s_i, a_i)$$

Note: There could be more than one policy with similar expected feature as π^* (or the data set).

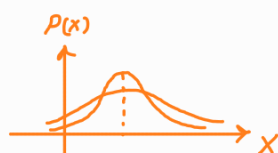
Occam's Razor $\rightarrow$ Regularization $\rightarrow$ Maximum Entropy Principle

Entropy: The entropy of a random variable x (or its probability distribution p) is a measure of its average uncertainty.

For discrete random variables, the entropy is defined as

$$\begin{aligned} \mathcal{H}(x) &= \mathbb{E}_{x \sim p} [-\log p(x)] \\ &= -\sum_{x \in \mathcal{X}} p(x) \log p(x) \geq 0. \end{aligned}$$

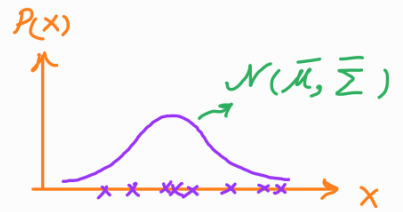
Maximum Entropy Principle: Suppose we are looking for a distribution among a set of distributions that matches a data set. If there are more than one distribution that fit the data, the maximum entropy principle pick the distribution with the highest uncertainty.



most general information

Example: Suppose we have n data points

$$\longrightarrow \begin{cases} \text{sample average } \bar{\mu} \\ \text{sample variance } \bar{\Sigma} \end{cases}$$



maximize
entropy

$$\max_{P_x} \mathcal{H}(P_x)$$

moment-
matching
constraints

$$\text{subject to } \begin{cases} \mathbb{E}_{x \sim P_x} [x] = \bar{\mu} \\ \mathbb{E}_{x \sim P_x} [xx^T] = \bar{\mu}\bar{\mu}^T + \bar{\Sigma} \end{cases}$$

$$\Rightarrow P_x^* = \mathcal{N}(\bar{\mu}, \bar{\Sigma})$$

Maximum Entropy IRL: Among the set of policies with matching expected feature, Max Ent-IRL aims to find the one with the highest entropy, i.e.,

$$\begin{aligned} & \max_{\pi \in \Pi} \mathcal{H}(\pi) \\ & \text{subject to } \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\pi_0}^{\pi^*}} [\phi(s,a)] = \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\pi_0}^{\pi}} [\phi(s,a)], \end{aligned}$$

where

$$\begin{aligned} \mathcal{H}(\pi) &= \mathbb{E}_{s \sim \bar{p}_{\pi_0}^{\pi}} \left[\mathbb{E}_{a \sim \pi(\cdot|s)} [-\log \pi(a|s)] \right] \\ &= - \mathbb{E}_{(s,a) \sim \bar{\gamma}_{\pi_0}^{\pi}} [\log \pi(a|s)] \end{aligned}$$

→

$$\hat{\pi} = \arg \min_{\pi} \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\log \pi(a|s)]$$

$$\text{subject to } \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi^*}} [\phi(s,a)] = \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\phi(s,a)]$$

This constrained optimization problem can be cast as

$$\min_{\pi \in \Pi} \max_{w \in \mathbb{R}^k} \left[\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\log \pi(a|s)] + w^T \left(\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi^*}} [\phi(s,a)] - \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\phi(s,a)] \right) \right]$$

by writing the Lagrange formulation.

In practice, we often solve the dual formulation:

$$\max_{w \in \mathbb{R}^k} \min_{\pi \in \Pi} \left[\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\log \pi(a|s)] + w^T \left(\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi^*}} [\phi(s,a)] - \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\phi(s,a)] \right) \right]$$

To solve this problem, we use an iterative constrained optimization algorithm.

Max Ent IRL

- Initialize the Lagrange multipliers w_0

- For $t=0, 1, 2, \dots, T-1$:

- Update policy

solved using
soft value/policy iteration

$$\pi_t = \arg \min_{\pi \in \Pi} \left[\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\log \pi(a|s)] - w_t^T \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\phi(s,a)] \right]$$

- Update Lagrange multipliers

gradient ascent

$$w_{k+1} = w_k + \eta \left(\mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi^*}} [\phi(s,a)] - \mathbb{E}_{(s,a) \sim \bar{\nu}_{\mu_0}^{\pi}} [\phi(s,a)] \right)$$

- Return $\bar{\pi} = \mathcal{U}(\pi_0, \pi_1, \pi_2, \dots, \pi_{T-1})$

uniform distribution